COMPARISON DIAGRAM OF RINGED TOPOI BETWEEN THE BIG FPPF SIMPLICIAL SITE AND THE SMALL ETALE SITE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the closing 'We omit the verification that the diagram is commutative (as a diagram of ringed topoi - we already know it is commutative as a diagram of topoi)' (source lines 7536-7541) covers only the lift of the already-proven topos commutativity to the ringed level: the underlying square commutes by the cited comparison lemma, and each morphism's ringed structure was individually established in-span (h_*O_{big,total} = O_U, the a_{U_n} and a_X from the cited More-on-Cohomology section, a_{fppf} from the hypercovering-modules section), so the ringed square is the compatibility of these constructed structures; the author's footnote on the provenance of u is preserved verbatim.

Editorial note. The label lemma-compare-simplicial-objects-fppf-etale-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The underlying comparison diagram of topoi. Same chapter, \label{lemma-compare-simplicial-objects-fppf-etale}. Lines 7506-7508: the topos-level diagram and the functor u are established in its proof, via lemma-morphism-augmentation-simplicial-sites (footnote).
- Structure sheaves of the simplicial algebraic space and the big site. Same chapter, Sections \label{section-simplicial-algebraic-spaces} and \label{section-hypercovering-modules}. Lines 7509-7516: O_U, O_X, O_{big,total}, O_{big} and a_{fppf} as a morphism of ringed topoi.
- The components as morphisms of ringed topoi. More on Cohomology of Spaces chapter, Section \label{spaces-more-cohomology-section-fppf-etale-modules}. Lines 7517-7520: h_n = a_{U_n} and h_{-1} = a_X are morphisms of ringed topoi.
- Morphisms of ringed simplicial sites and of ringed topoi. Same chapter, Remark \label{remark-morphism-simplicial-sites-modules} and \label{lemma-morphism-simplicial-sites-modules}. Lines 7524-7529: h_*O_{big,total} = O_U endows h with the ringed-simplicial-site structure, and the cited lemma produces h as a morphism of ringed topoi.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0DH5: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7233–7233. Prerequisite pointer invoked at proof line(s) 7508; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D99: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1402–1402. Prerequisite pointer invoked at proof line(s) 7527; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1604–1604. Prerequisite pointer invoked at proof line(s) 7534; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DGZ: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1588–1588. Prerequisite pointer invoked at proof line(s) 7532; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DAA: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5186–5186. Prerequisite pointer invoked at proof line(s) 7501; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7503; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DGM: sources/stacks-chow/tags.txt, source lines 16877–16877. Prerequisite pointer invoked at proof line(s) 7521; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. There is a commutative diagram*

$$\begin{array}{ccc} (Sh((Spaces/U)_{fppf,total}), \mathcal{O}_{big,total}) & \xrightarrow{h} & (Sh(U_{étale}), \mathcal{O}_U) \\ \downarrow{\scriptstyle a_{fppf}} & & \downarrow{\scriptstyle a} \\ (Sh((Spaces/X)_{fppf}), \mathcal{O}_{big}) & \xrightarrow{h_{-1}} & (Sh(X_{étale}), \mathcal{O}_X) \end{array}$$

of ringed topoi where the left vertical arrow is defined in Section 0DAA and the right vertical arrow is defined in Section 0DE7.

Proof. For the underlying diagram of topoi we refer to the discussion in the proof of Lemma 0DH5. The sheaf $\mathcal{O}_U$ is the structure sheaf of the simplicial algebraic space U as defined in Section 0DE7. The sheaf $\mathcal{O}_X$ is the usual structure sheaf of the algebraic space X. The sheaves of rings $\mathcal{O}_{big,total}$ and $\mathcal{O}_{big}$ come from the structure sheaf on $(\mathit{Spaces}/S)_{fppf}$ in the manner explained in Section 0DAA which also constructs a_{fppf} as a morphism of ringed topoi. The component morphisms $h_n = a_{U_n}$ and $h_{-1} = a_X$ are morphisms of ringed topoi by More on Cohomology of Spaces, Section 0DGM. Finally, since the continuous functor $u : U_{spaces,étale} \to (\mathit{Spaces}/U)_{fppf,total}$ used to define h^1 is given by $V/U_n \mapsto V/U_n$ we see that $h_*\mathcal{O}_{big,total} = \mathcal{O}_U$ which is how we endow h with the structure of a morphism of ringed simplicial sites as in Remark 0DGZ. Then we obtain h as a morphism of ringed topoi by Lemma 0DH0. Please observe that the morphisms h_n indeed agree with the morphisms a_{U_n} described above. We omit the verification that the diagram is commutative (as a diagram of ringed topoi – we already know it is commutative as a diagram of topoi). □

[1]This happened in the proof of Lemma 0DH5 via an application of Lemma 0D99.